

Yuchen Guo

Ph.D. Candidate in Physics, Tsinghua University | guo-yc23@mails.tsinghua.edu.cn | +86 199 4608 1290
github.com/yuchenguommm | LinkedIn | Google Scholar
Based in Beijing; in Hong Kong from September 2026 to January 2027

EDUCATION

Tsinghua University — Ph.D. in Physics	2023 – 2028 (expected)
Tsinghua University — B.Sc. in Physics. GPA 3.92/4.0, Rank 1 of 120	2019 – 2023
The Chinese University of Hong Kong — Junior Research Assistant	Sep 2026 – Jan 2027
Shanghai High School — olympiad track in mathematics and physics	2016 – 2019

COMPETITIONS AND HONOURS

- **Chinese Physics Olympiad — Gold Medal and National Team (国集), 2018.** Top 50 nationally; admitted to Tsinghua ahead of the standard cycle.
- **Tsinghua University Special Scholarship (清华特奖), 2022.** Highest student honour at Tsinghua; ten recipients university-wide per year.
- National Scholarship for Ph.D. Students, 2025 · NSFC Young Student Basic Research Project (independent grant), 2025
- Outstanding Graduate and Outstanding Bachelor Thesis, Tsinghua University and Beijing, 2023

TECHNICAL SKILLS

Languages: Python (expert)
Numerical and optimisation: large-scale linear algebra, tensor contraction, MILP (Gurobi), Monte Carlo, stochastic optimisation
Machine learning and data: PyTorch, NumPy/SciPy, evolutionary search

EXPERIENCE

Optiver — Quantitative Research Intern (summer 2027, offer accepted) 2026 – 2027
Offer following PhD QuantFocus, a selective programme covering mental arithmetic, options and contract pricing, market-making strategy design, and factor research.

- **Placed first in the trading-strategy competition** (timed build, ranked against the full cohort across nine parameter regimes): ~30% ahead of second place and the only entry profitable in every regime.

FieldQuantum — Founding Partner and Head of Quantum Algorithms Feb 2026 – Aug 2026
Venture-backed startup, approx. 50 staff; ¥245M raised, backed by Sequoia China and InnoAngel.

- Built and led the quantum algorithm team; owned technical roadmap, hiring, and delivery.
- Shipped **fieldqkit**, an open-source Python SDK for quantum computation.
- Led a 12,000-atom molecular simulation programme.
- Authored the company whitepaper on real-time quantum error correction (arXiv:2605.30765); designed evolutionary search over variational ansätze using AlphaEvolve.

Tsinghua University — Doctoral Researcher 2023 – present
Tensor-network methods for quantum computation, open quantum systems, and topological matter; analytical derivation combined with large-scale numerical simulation.

RESEARCH OUTPUT

Twelve first-authored papers in *Phys. Rev. X*, *Phys. Rev. Lett.*, *PRX Quantum*, *Rep. Prog. Phys.*, and *npj Quantum Information*.
Selected: *Phys. Rev. Lett.* **137**, 110401 (2026) · *Phys. Rev. X* **15**, 021060 (2025) · *Phys. Rev. Lett.* **130**, 220401 (2023) · *PRX Quantum* **3**, 040313 (2022) · *Rep. Prog. Phys.* **88**, 118001 (2025)
Talks: APS Global Physics Summit (2025), Asia-Pacific Physics Conference (2025).

OTHER

First and corresponding author of a cryptography proofs paper on the Simon dihedral coset problem (ePrint 2026/1714). Physics olympiad coach for several years. Writes a long-running essay column on science and technology. Classical pianist; performs each semester.